

Part 1: Problem-Solving Process

Step 1: Understand and Define the Problem (Analyse)

Problem Statement:

The animal shelter requires a **low-cost automated pet feeder system** that dispenses food to cats and dogs at scheduled times, monitors whether food is eaten, and alerts staff if there are issues (e.g., food not dispensed, food not eaten, food bin empty).

Features:

- Scheduled feeding (e.g., 8AM, 6PM).
- Dispensing mechanism (servo motor).
- Monitoring consumption (weight sensor in bowl).
- Alerts to staff (LED/sound/buzzer/notification).

Inputs:

- Feeding schedule (real-time clock).
- Food container level sensor.
- Weight sensor under bowl (to check if food is consumed).

Outputs:

- Servo motor rotates to dispense food.
- System alert (if no food eaten or bin empty).

Assumptions & Limitations:

- One type of dry pet food.
- Pets eat within 10 minutes of dispensing.
- Limited hardware resources (memory, simple sensors).
- Small shelter use only (not commercial scale).

Simple Block Diagram (sketch idea):

- Input (Clock, Food level sensor, Bowl weight sensor) → Processing Unit (Logic) → Output (Servo motor, Alert system).